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List of Formulas

→ High Side injection = $f_{LO} > f_{RF}$

→ Low Side injection = $f_{LO} < f_{RF}$

→ $f_{LO} > f_{RF}$ then

$f_{LO} = f_{RF} + f_{IF}$

or $\boxed{f_{IF} = f_{LO} - f_{RF}}$

→ $f_{LO} < f_{RF}$ then

$f_{LO} = f_{RF} - f_{IF}$

$f_{IF} = f_{RF} - f_{LO}$

→ Image frequency Rejection Ratio (IFRR)

$IFRR = \sqrt{1 + Q^2 \rho^2}$

→ $\rho = \left(\frac{f_{im}}{f_{RF}}\right) - \left(\frac{f_{RF}}{f_{im}}\right)$

→ f_{image} (f_{im}) $= f_{RF} + 2 f_{IF}$

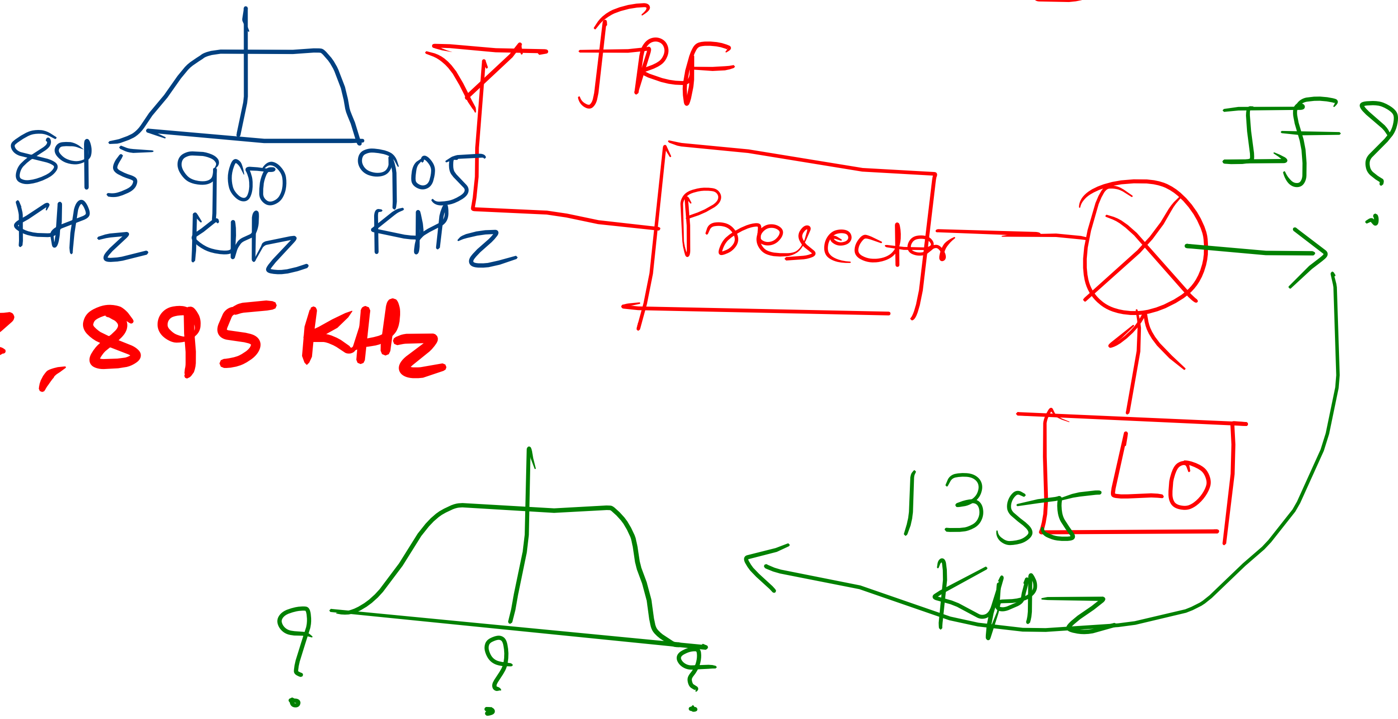
Example 1

$f_{LO} > f_{RF}$

For an AM Superheterodyne receiver that uses high-side injection and has LO of 1355 KHz, determine the IF carrier, upper side frequency, lower side frequency for and RF wave which is made up of carrier and upper and lower side frequency of 900 KHz, 905 KHz, and 896 KHz respectively.

$f_{LO} = 1355\ KHz$

$f_{RF} = 900\ KHz, 905\ KHz, 895\ KHz$



Example 1

For an AM Superheterodyne receiver that uses high-side injection and has LO of 1355 KHz, determine the IF carrier, upper side frequency, lower side frequency for and RF wave which is made up of carrier and upper and lower side frequency of 900 KHz, 905 KHz, and 896 KHz respectively.

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$$F_{IF} = F_{LO} - F_{RF}$$

$$F_{IF} = 1355 - 900\,KHz$$

$$\boxed{F_{IF} = 455\,KHz}$$

$$F_{IF}(upper) = F_{LO} - F_{RF}(upper)$$

$$= 1355 - 905\,KHz$$

$$F_{IF}(\cancel{upper}\ \text{Lower}) = 450\,KHz$$

$$F_{IF}(lower) = F_{LO} - F_{RF}(lower)$$

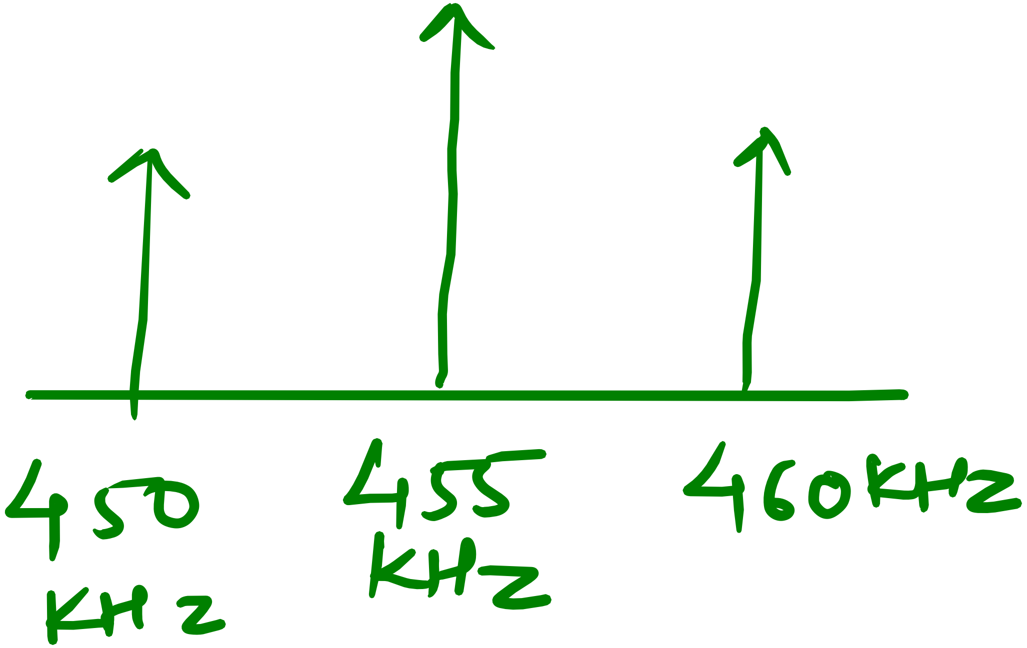
$$= 1355 - 895$$

$$F_{IF}(\cancel{lower}\ \text{upper}) = 460\,KHz$$

Example 1

For an AM Superheterodyne receiver that uses high-side injection and has LO of 1355 KHz, determine the IF carrier, upper side frequency, lower side frequency for and RF wave which is made up of carrier and upper and lower side frequency of 900 KHz, 905 KHz, and 896 KHz respectively.

Intermediate frequency



Example 2

For an AM broadcast band SH receiver with IF, RF and LO frequencies of 455 KHz, 600 KHz and 1055 KHz respectively, determine (i) Image frequency (ii) IFRR for a pre-selector of Q=100

$f_{IF} = 455\ kHz$

$f_{RF} = 600\ kHz$

$f_{LO} = 1055\ kHz$

$f_{image} = f_{RF} + 2f_{IF}$

$= 600 + 2(455)$

$\boxed{f_{image} = 1510\ kHz}$

Example 2

For an AM broadcast band SH receiver with IF, RF and LO frequencies of 455 KHz, 600 KHz and 1055 KHz respectively, determine (i) Image frequency (ii) IFRR for a pre-selector of Q=100

$$IFRR = \sqrt{1+Q^2\rho^2}$$

$$\rho = \left(\frac{f_{im}}{f_{RF}}\right) - \left(\frac{f_{RF}}{f_{im}}\right)$$

$$\boxed{\rho = 2.119}$$

$$IFRR = \sqrt{1+(100)^2(2.11)^2}$$

$$IFRR = 211.9 \approx \boxed{212}$$

Example 3

For a AM receiver having NO RF amplifier, the loaded Q of an antenna coupling circuit is 100. If the IF is 455 KHz, determine (i) IF and its rejection ratio for RF of 1000 KHz (ii) IF and its rejection ratio for RF of 25 MHz. Comment on results.

$Q = 100$

$f_{IF} = 455 KHz$

(i) IFRR for $f_{RF} = 1000 KHz$

(ii) IFRR for $f_{RF} = 25 MHz$

(i) $f_{image} = f_{RF} + 2f_{IF}$

$f_{image} = 1910 KHz$

$\rho = 1.38$

$IFRR = 138.6$

Example 3

For a AM receiver having NO RF amplifier, the loaded Q of an antenna coupling circuit is 100. If the IF is 455 KHz, determine (i) IF and its rejection ratio for RF of 1000 KHz (ii) IF and its rejection ratio for RF of 25 MHz

(ii) $F_{RF} = 25\ MHz$

$F_{image} = 25.910\ MHz$

$\rho = 0.071$

$IFRR = 7.21$

$F_{RF} = 1000\ KHz$

$IFRR = 138.6$ — 1

$F_{RF} = 25\ MHz$

$IFRR = 7.21$ — 2

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Example 4

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For a citizen band receiver using high-side injection with an RF carrier of 27 MHz and IF of 455 KHz, determine (i) LO (ii) Image frequency (iii) IFRR for a pre-selector Q of 100 (iv) Preselector Q required to achieve the same IFRR as that achieved for an RF carrier of 600 KHz in example (2)

$\rightarrow 212$

$f_{RF} = 27\text{ MHz}$

$f_{IF} = 455\text{ KHz}$

$f_{LO} = ?$

$F_{image} = ?$

$f_{LO} = f_{RF} + f_{IF}$

$= 27 \times 10^6 + 455 \times 10^3$

$\boxed{F_{LO} = 27.455\text{ MHz}}$

$f_{image} = 27.91\text{ MHz}$

$\rho = 0.066$ $\boxed{IFRR = 6.7}$

Example 4

For a citizen band receiver using high-side injection with an RF carrier of 27 MHz and IF of 455 KHz, determine (i) LO (ii) Image frequency (iii) IFRR for a pre-selector Q of 100 (iv) Preselector Q required to achieve the same IFRR as that achieved for an RF carrier of 600 KHz in example (2)

(iv)

$F_{RF} = 600\ KHz$

$\rho = 0.066$

$IFRR = \sqrt{1+Q^2\rho^2}$

$212 = \sqrt{1+Q^2\rho^2}$

$\boxed{Q = 3212}$

Example 4

For a citizen band receiver using high-side injection with an RF carrier of 27 MHz and IF of 455 KHz, determine (i) LO (ii) Image frequency (iii) IFRR for a pre-selector Q of 100 (iv) Preselector Q required to achieve the same IFRR as that achieved for an RF carrier of 600 KHz in example (2)

LO = 27.455 MHz

f_{image} = 27.91 MHz

IFRR(Q = 100) = 6.7

Required Q = 3200 → Impractical

Poor rejection

At high RF (27 MHz) → image is very close

Example 4

For a citizen band receiver using high-side injection with an RF carrier of 27 MHz and IF of 455 KHz, determine (i) LO (ii) Image frequency (iii) IFRR for a pre-selector Q of 100 (iv) Preselector Q required to achieve the same IFRR as that achieved for an RF carrier of 600 KHz in example (2)

Hence real receiver uses

→ RF amplifier stages

→ Higher IF (or double conversion receivers)

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Example 5

- **A superheterodyne receiver uses low-side injection to receive a signal at 1.2 MHz. The IF is 455 kHz.**

Determine:
(i) Local oscillator frequency
(ii) Image frequency
(iii) IFRR if preselector Q = 80
(iv) Comment on image rejection quality

$$f_{LO} = f_{RF} - f_{IF}$$

$$f_{LO} = 1200 - 455$$

$$\boxed{f_{LO} = 745\ \text{KHz}}$$

$$f_{image} = f_{RF} - 2f_{IF}$$

$$= 1200 - 910$$

$$\boxed{f_{image} = 290\ \text{KHz}}$$

Example 5

- **A superheterodyne receiver uses low-side injection to receive a signal at 1.2 MHz. The IF is 455 kHz.**

Determine:
(i) Local oscillator frequency
(ii) Image frequency
(iii) IFRR if preselector Q = 80
(iv) Comment on image rejection quality

$$\rho = \left(\frac{f_{im}}{f_{RF}}\right) - \left(\frac{f_{RF}}{f_{im}}\right) = -3.895$$

$$IFRR = 311.6$$

$$IFRR \text{ in dB} = 20 \log(311.6) = 49.9 \text{ dB}$$

Example 5

- **A superheterodyne receiver uses low-side injection to receive a signal at 1.2 MHz. The IF is 455 kHz.**

Determine:
(i) Local oscillator frequency
(ii) Image frequency
(iii) IFRR if preselector Q = 80
(iv) Comment on image rejection quality

→ Image frequency (290 kHz) is far from RF (1.2 MHz)

→ This leads to very high IFRR (~50 dB)

→ Hence, good image rejection.